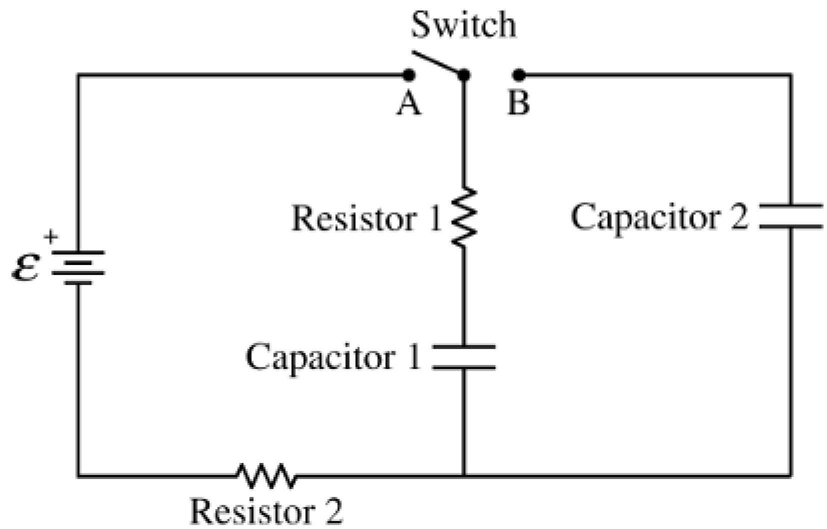
Station 1 - 4 pts

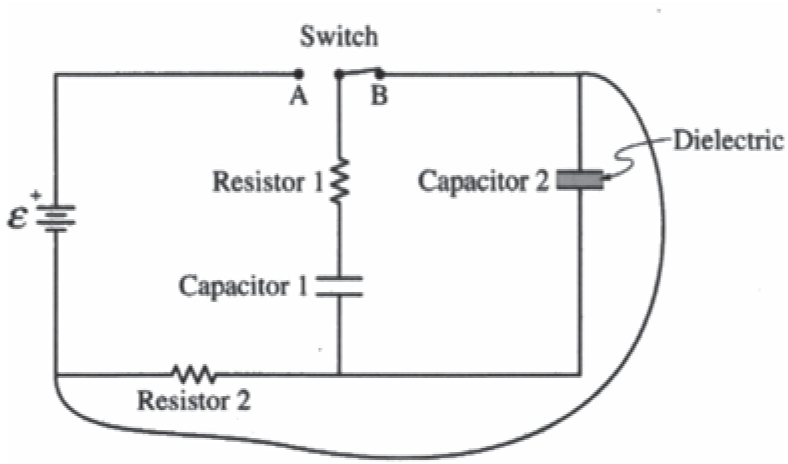


The circuit shown consists of an ideal battery of emf ε, resistors 1 and 2 each with resistance R, capacitors 1 and 2 each with capacitance C, and a switch. The switch is initially open, and both capacitors are unchanged.

At time $t = 0$, the switch is closed to Position A. A long time after the switch is closed to Position A, the total charge on the positive plate of Capacitor 1 is Q0 and Capacitor 2 is uncharged. At time t1, the switch is closed to Position B.

1. On the axes shown in your exam booklet, sketch graphs of the current IR in Resistor 1 and the energy UC stored in Capacitor 1 as functions of time t from time t = 0 until steady-state conditions are nearly reached. ***TB 1* (1 pt)**
2. Immediately after t1, is the direction of the current in Resistor 1 directed up, directed down, or is there no current? Briefly justify your answer. **(1 pt)**
3. Determine an expression for the total charge on the positive plate of Capacitor 2 a long time after t1. Express your answer in terms of Q0 and fundamental constants, as appropriate. **(1 pt)**
4. Derive an expression for the total energy dissipated by Resistor 1 immediately after time t1 until new steady-state conditions have been reached. Express your answer in terms of C, Q0, and fundamental constants, as appropriate. **(1 pt)**

<u>Station 2 - 3 pts</u>



The circuit shown consists of an ideal battery of emf ε, resistors 1 and 2 each with resistance R, capacitors 1 and 2 each with capacitance C, and a switch. The switch is initially open, and both capacitors are unchanged. At time t = 0, the switch is closed to Position A. At time t1, the switch is closed to Position B.

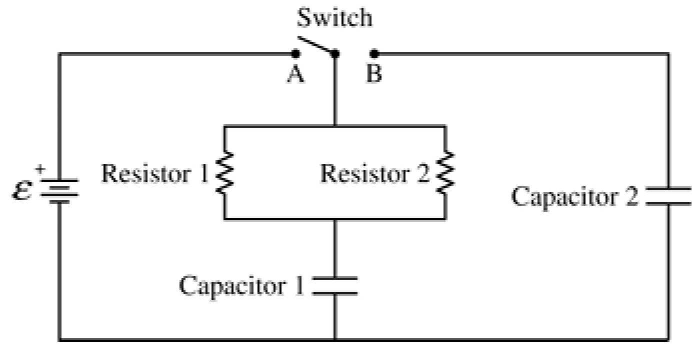
With the switch then remaining closed to Position B, a dielectric material with dielectric constant k = 2 is inserted between the plates of Capacitor 2.

5. Determine the charge on the positive plate of Capacitor 2 a long time after the dielectric has been inserted. Express your answer in terms of Q0 and fundamental constants, as appropriate. **(1 pt)**

6. Derive an expression for the current in Resistor 2 immediately after the wire is connected to the circuit. Express your answers to the following two questions in terms of R, C, Q0, and fundamental constants, as appropriate. ***TB 2* (1 pt)**

7. Determine the current in Resistor 2 a long time after the wire is connected to the circuit. Express your answers to the following two questions in terms of R, C, Q0, and fundamental constants, as appropriate. **(1 pt)**

Station 3 - 4 pts



The circuit shown consists of a battery of emf Ɛ, resistors 1 and 2 each with resistance R, capacitors 1 and 2 with capacitances C and 2C, respectively, and a switch. The switch is initially open, and both capacitors are uncharged.

At time t = 0, the switch is closed to Position A. A long time after the switch is closed to Position A, the charge on the positive plate of Capacitor 1 is Q0 and Capacitor 2 is uncharged. At time t1, the switch is closed to Position B.

8. Derive an expression for the total energy E_R dissipated by resistors 1 and 2 from immediately after time t_1 until new steady-state conditions have been reached. Express your answer in terms of C, Q_0, and fundamental constants, as appropriate. Simplify your answers accordingly. **(1 pt)**

   With the switch still closed to Position B, the parallel plates of Capacitor 2 are moved so that the separation distance increases by a factor of 2.

9. Determine the ratio of $\frac{U_2}{U_1}$ of the energy U2 stored in Capacitor 2 to the energy U1 stored in Capacitor 1 a long time after the plates of Capacitor 2 have been moved. **(1 pt)**

   With the capacitors still charged, the switch is now closed to Position A.

10. Derive an expression for the current I0 from the battery immediately after the switch is closed to Position A. Express your answers in terms of R, C, Q0, and fundamental constants, as appropriate. **(1 pt)**

11. Determine the current I∞ from the battery a long time after the switch is closed to Position A. Express your answers in terms of R, C, Q0, and fundamental constants, as appropriate. **(1 pt)**

Station 4 - 21 pts

12. Write the letter matching the SI units with the quantity **(5 points - 1 point each)**

___ Coulomb	A. $\frac{m^2 kg}{s^3 A}$
___ Watt	B. $s * A$
___ Volt	C. $\frac{s^4 A^2}{m^2 kg}$
___ Ohm	D. $\frac{m^2 kg}{s^3 A^2}$
___ Farad	E. $\frac{m^2 kg}{s^3}$

13. Match each name with its definition. **(6 points - 1 pt each)**

___ Kirchhoff's Voltage Law	A. Relates voltage, current and resistance
___ Kirchhoff's Current Law	B. The sum of voltages on a closed loop is 0
___ Mesh Current Analysis	C. The sum of currents entering a node is 0
___ Norton Equivalent Circuit	D. Reduction of a circuit to a current and resistor
___ Thévenin Equivalent Circuit	E. Applies (B) for multiple loops
___ Ohm's Law	F. Reduction of a circuit to a voltage source and resistor

14. True or False - Answer the following with T or F. **(10 points - 1 pt each)**

___ For a parallel-plate capacitor connected to a fixed voltage source, inserting a dielectric material with a high dielectric constant will always decrease the energy stored in the capacitor. **(1 pt)**

___ In a simple DC circuit with a single resistor connected to a battery, replacing the resistor with one of half the resistance will result in the battery delivering four times the power. **(1 pt)**

___ According to Kirchhoff's Current Law, the sum of currents entering a node must be zero, which implies that charge cannot accumulate at any point in a circuit under steady-state DC conditions. **(1 pt)**

___ The magnetic field lines produced by a straight, current-carrying wire form closed loops that are concentric circles around the wire, as determined by the right-hand rule. **(1 pt)**

___ A Ground Fault Circuit Interrupter (GFCI) operates by comparing the current in the "hot" and "neutral" wires; a difference as small as 5 mA will trigger it to open the circuit. **(1 pt)**

___ The primary hazard associated with common household AC electricity, as opposed to DC of the same voltage, is its higher peak voltage, which is approximately 170 V for a 120 V RMS system. **(1 pt)**

___ Michael Faraday's law of induction quantitatively describes how a changing magnetic field can induce an electromotive force (EMF), while the work of André-Marie Ampère is central to understanding the magnetic field generated by an electric current. **(1 pt)**

___ For an ideal transformer, if the primary coil has twice as many turns as the secondary coil, the voltage measured across the secondary will be half the voltage applied to the primary. **(1 pt)**

___ A circuit breaker is designed to protect wiring and appliances from current overload, while a fuse provides protection from both overload and ground faults. **(1 pt)**

___ In a series circuit, the resistor with the largest resistance will always dissipate the most power. **(1 pt)**

Station 5 - 18pts

15. Which battery type typically has the highest energy density? **(1 pt)**
    a. Lead-acid
    b. Nickel-cadmium
    c. Lithium-ion
    d. Alkaline

16. Which safety feature is most critical for DC battery systems? ***TB3* (1 pt)**
    a. GFCI protection
    b. Proper fusing for overcurrent protection
    c. Isolation transformers
    d. Automatic voltage regulation

17. The peak voltage of a 120V RMS AC circuit is approximately: **(1 pt)**
    a. 84.8V
    b. 120V
    c. 170V
    d. 240V

18. The frequency of AC power in North America is: ***TB4* (1 pt)**
    a. 50 Hz
    b. 60 Hz
    c. 100 Hz
    d. 120 Hz

19. Two identical conducting spheres, one with charge +Q and the other neutral, are briefly touched together and separated. What is the final charge on each sphere? **(1 pt)**
    a. +Q on both
    b. +Q/2 on both
    c. +Q on one, zero on the other
    d. +Q/4 on both

20. If the distance between two point charges is tripled, the force between them becomes: **(1 pt)**
    a. 1/9 of the original
    b. 1/3 of the original
    c. 3 times the original
    d. 9 times the original

21. The resistance of a conductor increases with temperature because: **(1 pt)**
    a. Electron density decreases
    b. Electron charge decreases

c. Lattice vibrations increase, causing more electron scattering
d. The material expands, increasing length

22. The energy consumed by a 1.5kW heater running for 2 hours is: **(1 pt)**
    a. 3 kWh
    b. 750 Wh
    c. 3,000 J
    d. 1.5 kWh

23. In an electric motor, the function of the commutator is to: **(1 pt)**
    a. Convert AC to DC
    b. Reverse current direction in the coils as they rotate
    c. Increase magnetic field strength
    d. Reduce sparking

24. A step-up transformer: ***TB5* (1 pt)**
    a. Increases voltage and decreases current
    b. Increases voltage and increases current
    c. Decreases voltage and increases current
    d. Decreases voltage and decreases current

25. The National Electrical Code requires GFCIs in: **(1 pt)**
    a. All kitchen countertop outlets
    b. All bathroom outlets
    c. All garages and outdoor areas outlets
    d. All the above
    e. None of the above

26. A three-way switch is used to: ***TB6* (1 pt)**
    a. Control a light from three locations
    b. Control a light from two locations
    c. Provide three different light levels
    d. Switch between three different circuits

27. Three 10Ω resistors in parallel have an equivalent resistance of: **(1 pt)**
    a. 30Ω
    b. 10Ω
    c. 3.33Ω
    d. 0.3Ω

28. In a parallel circuit, if one branch opens: **(1 pt)**
    a. All currents decrease
    b. Total current decreases
    c. Total current increases

d. Voltage drops to zero

29. At a junction with three wires, if I_1 = 2A entering and I_2 = 1A leaving, then I_3 must be: **(1 pt)**
    a. 1A entering
    b. 1A leaving
    c. 3A entering
    d. 3A leaving

30. A node in a circuit has 2A, 3A, and I entering, and 6A leaving. What is I? **(1 pt)**
    a. 1A entering
    b. 1A leaving
    c. 11A entering
    d. 11A leaving

31. In a purely inductive AC circuit, how does the instantaneous current (I) relate to the instantaneous voltage (V)? **(1 pt)**
    a. Voltage lags the current by 90°.
    b. Current leads the voltage by 90°.
    c. Current and voltage are in phase (ϕ=0).
    d. Voltage leads the current by 90°.

32. According to Lorentz Force Law, what is true? Select all that apply **(1 pt)**
    a. Force is applied on all charged particles by both magnetic and electric fields
    b. If a charge isn't moving there is no magnetic force
    c. The Force of the Magnetic Field is equal to Fmag = qv x B and is in the direction diagonal to the magnetic field and the moving charge
    d. The Force of the Electric Field is not equal to Felect = qE but is in the direction of the electric field

<u>Station 6 - Practical - 7 pts</u>

There is a circuit built using standard wire-lead resistors connected to a 9-volt source. Take note of the circuit layout and use the multimeter provided to make any necessary measurements. **Do not alter the power supply, and do not remove any resistors from the circuit.** Include units, if possible, in all of your answers

33. Voltage Drop - R1: _______ ***TB7* (1 pt)**
34. Current through - R2: _______ **(1 pt)**
35. Make one extra measurement necessary to determine all 3 individual resistances, given the values you found above (circle your response):
    The *(voltage/current)* across *(R1/R2/R3)*: _____ **(3 pts)**

36. What is the total resistance of the system? _____ ***TB8* (1 pt)**
37. Neatly draw a circuit diagram of this circuit, including all values for resistor(s) and the voltage source(s). **(1 pt)**

Station 7 - Practical - 9 pts

There are three circuits set up on a breadboard, each with a white LED bulb and some combination of resistors. Connect the power supply (already set at 6 V) to this circuit at the indicated points. Do not remove the resistors or bulbs or alter the power supply setting.

These are the three circuit setups:
Setup 1 100Ω only
Setup 2 220Ω only
Setup 3 100Ω and 330Ω in series

38. Use the multimeter provided to make all measurements necessary to complete the following chart. **(9 pts - 1 pt each square)**

	Current	Voltage Drop Across the Bulb	Power Dissipated in the Bulb
Setup 1			
Setup 2			
Setup 3			

Station 8 - Practical - 17 pts

39. Find the electrical potential of the battery pack number 1 and battery pack number 2. **(2 pts)**

40. Set up a circuit with 3 resistors in SERIES using the given color combos and the NUMBER 2 battery pack. **(9 pts - 1 pt each square)**
    Use your multimeter to find the voltage across each resistor.
    Use a method, without calculating, to determine the resistance
    Enter the values in the table in your booklet.

Resistor	Voltage	Current	Resistance
Orange, Orange, Brown, Gold			
Brown, Black, Black, Gold			
Brown, Black, Red, Gold			

41. Set up a circuit shown below using the NUMBER 1 battery pack.
    Use your multimeter to find the voltage and current across each resistor.
    Enter the values into the table in your booklet. **(6 pts - 1 pt each square)**

Resistor	Voltage	Current
Orange, Orange, Brown, Gold		
Brown, Black, Black, Gold		
Brown, Black, Red, Gold		